CBSE Test Paper 01

Chapter 11 Thermal Properties of Matter

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1. Molar Specific Heat capacity has units of **1**
   a. W/m.K
   b. J/mol.K
   c. J.ohm/sec.K^2
   d. J/kg.K

2. A hole is drilled in a copper sheet. The diameter of the hole is 4.24 cm at 27.0^oC. What is the change in the diameter of the hole when the sheet is heated to 227^oC? Coefficient of linear expansion of copper = $1.70 \times 10^{-5}\ K^{-1}$. **1**
   a. 1.24 x 10^{-2} cm
   b. 1.44 x 10^{-2} cm.
   c. 1.34 x 10^{-2} cm
   d. 1.54 x 10^{-2} cm

3. A student eats a dinner rated at 2000 Calories. He wishes to do an equivalent amount of work in the gymnasium by lifting a 50.0-kg barbell. How many times must he raise the barbell to expend this much energy? Assume that he raises the barbell 2.00 m each time he lifts it and that he regains no energy when he drops the barbell to the floor. **1**
   a. 8.14×10^3
   b. 8.74×10^3
   c. 8.54×10^3
   d. 8.34×10^3

4. A spray can containing a propellant gas at twice atmospheric pressure (202 kPa) and having a volume of 125 cm^3 is at 22^oC. It is then tossed into an open fire. When the temperature of the gas in the can reaches 195^oC, what is the pressure inside the can?

Assume any change in the volume of the can is negligible. **1**

a. 260 kPa
b. 300 kPa
c. 290 kPa
d. 320 kPa

5. A steel railroad track has a length of 30.000 m when the temperature is 0.0^{o}C. Suppose that the ends of the rail are rigidly clamped at 0.0^{o}C so that expansion is prevented. What is the thermal stress set up in the rail if its temperature is raised to 40.0^{o}C? $\alpha - steel$ = 11×10^{-6} ($^{\circ}$ C)-1 Y-steel = 20GPa **1**

   a. 8.4×10^{6} N/m^2
   b. 8.1×10^{7} N/m^2
   c. 9.1×10^{7} N/m^2
   d. 8.8×10^{6} N/m^2
6. Why the temperature above 1200°C cannot be measured accurately by a platinum resistance thermometer? **1**
7. Stainless steel cooking pans are preferred with extra copper bottom. Why? **1**
8. Each side of a cube increases by 0.01% on heating. How much is the area of its faces and volume increased? **1**
9. Calculate the heat of combustion of coil when 10 g of coal on burning raises the temperature of 2 litres of water from 20^{o} C to 55^{o} C. **2**
10. An iron ring of diameter 5.231 m is to be fixed on a wooden rim of diameter 5.243 m both initially at 27°C. To what temperature should the iron ring be heated so as to fit the rim? (Coefficient of linear expansion of iron is 1.2×10^{-5} K^{-1}) **2**
11. What do you mean by molar specific heat of a substance? What is its SI unit? **2**
12. A 'thermacole' icebox is a cheap and efficient method for storing small quantities of cooked food in summer in particular. A cubical icebox of side 30 cm has a thickness of 5.0 cm. If 4.0 kg of ice is put in the box, estimate the amount of ice remaining after 6 h. The outside temperature is 45 °C, and co-efficient of thermal conductivity of thermacole is 0.01 J s^{-1}m^{-1}K^{-1}. [Latent heat of melting of ice = 335×10^{3} Jkg^{-1}] **3**

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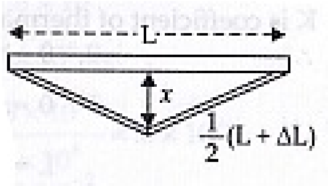
14. What is calorimetry? Briefly explain its principle. **3**
15. A rail track made of steel having length 10 m is clamped on a railway line at its two ends (Figure).



On a summer day due to rise in temperature by 20° C, it is deformed as shown in figure. Find *x* (displacement of the centre) if α_{steel}= 1.2 $\times$ 10^{-5} /°C. **5**